

Tribological and mechanical characterization of glass fiber polyamide composites under hydrothermal aging

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ABSTRACT

Polymer composites are increasingly used in applications where they are subjected to abrasive wear. The bulk of prior research has focused on the performance of polymers under sliding wear conditions and in some cases resulted in a contradiction, particularly when it comes to the effect of reinforcing fibers on the wear performance. Another omission is the absence of pre-conditioning or accelerated aging which is clearly an important factor from an application standpoint. This work aims to highlight the influence of hydrothermal aging on the tribological and mechanical performance of polyamide 6 (PA6) and glass fiber-reinforced polyamide 6 (PA6-GF). The friction and abrasion wear behavior of unconditioned and conditioned PA6 and PA6-GF composites are measured against silicon carbide grinding paper on a pin-on-disk tribometer in a demineralized water medium at 60 °C. Specimens are conditioned in demineralized water at room temperature (RT) 25, 40, and 60 °C to investigate the effects of moisture and temperature. The presence of up to 25 wt.% glass fiber (GF) in PA6-based composites exhibits an increasing trend in abrasion wear mass loss and a reduction in coefficient of friction (COF). Interestingly, the mass loss of specimens conditioned to saturation does not increase compared to that of unconditioned samples. The FTIR spectroscopy is used to study the polymer degradation during hydrothermal aging.

1. Introduction

The successful development and use of advanced engineering products hinge on a comprehensive understanding of wear. Polymers and their composite materials are being increasingly used in wear applications, replacing metals due to advantages such as low cost, low mass, good wear performance, reduced wear of the counter surface, and enhanced corrosion resistance. Much of the literature on the polymers' tribological behaviors has paid particular attention to sliding wear. However, in industrial applications, materials are exposed to more severe wear conditions, therefore more attention is needed for the analysis of the tribological behaviors of polymer composites under abrasion wear, and hydrothermal aging conditions [1,2].

Polymeric composites reinforced by fibers demonstrate excellent mechanical properties, however, their weak behavior towards hydrothermal aging conditions limits their application in aggressive environments. Hydrothermal aging conditions can affect the mechanical,

thermal, tribological, and physiochemical properties of polymers and polymer composites [3]. The water diffusion process in fiber-reinforced polymer (FRP) composites can take place through various modes. While the water spreads through the polymer matrix, fiber/ matrix interfaces seem to be a predominant pathway for water uptake through a network of microchannels established along the fiber/ matrix interface [4–8]. In addition, there have been some contradictions in the influence of fibers on the wear resistance of polymers' matrix in different wear modes [9–12]. Some reports demonstrated that FRPs had higher wear resistance in sliding mode, while fibers deteriorated the wear resistance of polymer composites in abrasion modes [9,10,12].

Polyamide 6 (PA6) known as Nylon 6 is one of the most extensively used high-performance engineering thermoplastics because of its excellent mechanical, thermal, extrusion-ability, mould-ability, good chemical resistivity, and wear resistance. PA6 is a suitable candidate for developing tribo-composites with low friction and wear [13–15]. However, PA6 is sensitive to moisture in the hydrothermal environment

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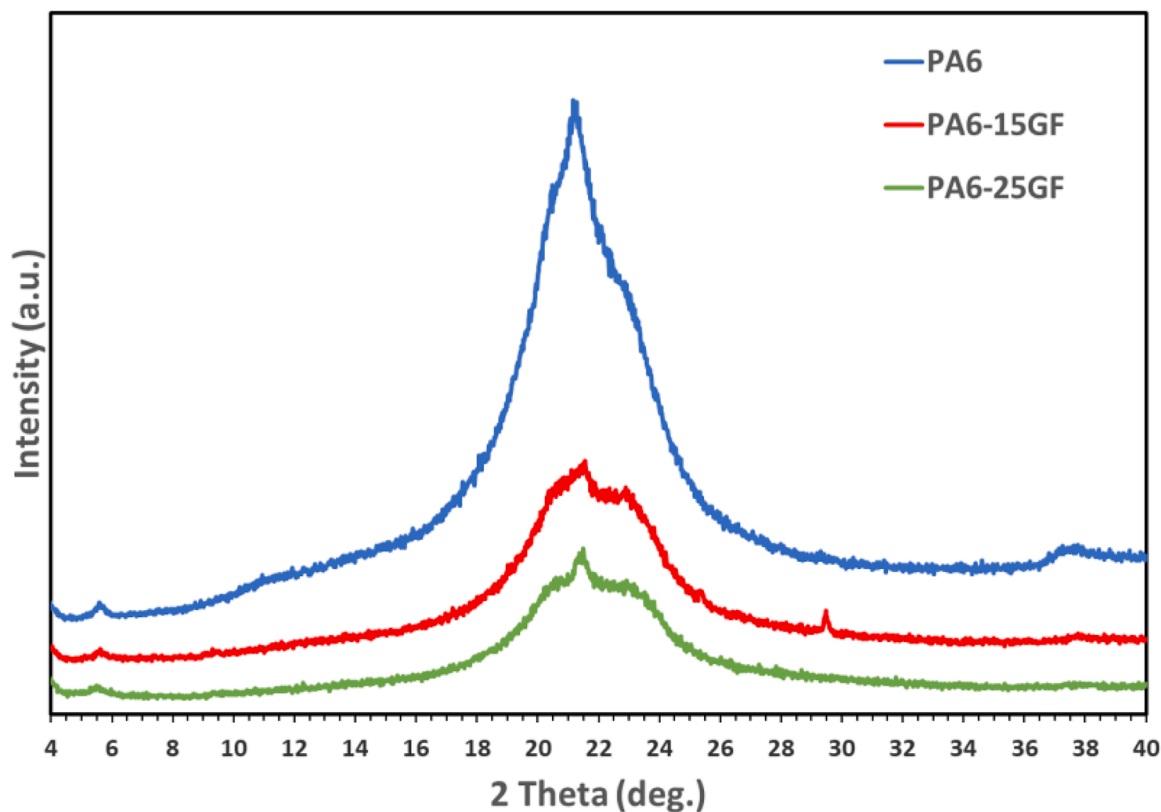


Fig. 1. XRD spectrum of the PA6, and composites of PA6-15GF and PA6-25GF.

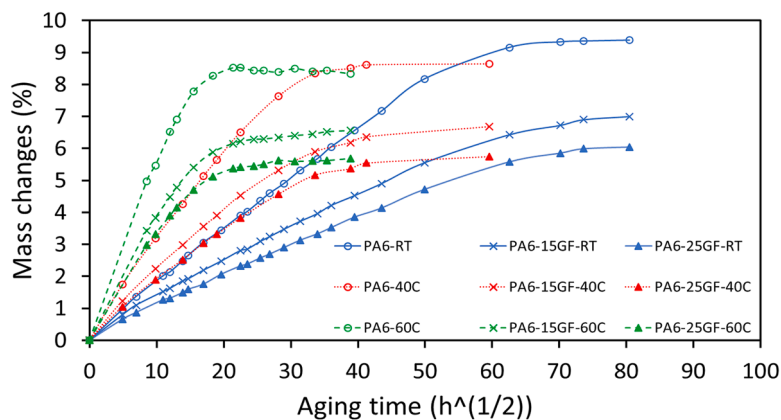


Fig. 2. Water absorption curves of PA6 and its PA6-GF composites at RT, 40 °C and 60 °C.

which affects its performance and durability [16]. Glass fiber-reinforced polymeric composites are commonly used in composite material fabrication. Glass fibers (GFs) are used as reinforcing agents in polymeric composites due to their excellent properties such as high strength, flexibility, stiffness, and chemical resistance [17–20]. Given these characteristics, it becomes imperative to investigate behavioral changes of PA6 composites under hydrothermal conditions. Ferreno et al. [21] modelled the influence of increasing water content on the mechanical properties of polyamide 6-glass fiber (PA6-GF) composites. Their analysis revealed that as water uptake increases, the toughness of composites can improve. Materials softening led to a decrease in Young's modulus and ultimate strength and an increase in strain at fracture. With increasing the water content, the brittle feature was replaced with considerable plastic deformation prior to fracture accompanied by notable debonding between the matrix and fibers. The effects of

hydrothermal aging on the mechanical properties of PA6 and PA6-GF composites have been investigated by Ksouri and Haddar [16]. Increasing conditioning temperature from 30 °C to 90 °C showed a significant loss in stiffness and strength of specimens caused by plasticization effect and/or interfacial debonding. The temperature effect was seen in PA6 samples more than in PA6-GF composites.

In this study, the tribological behavior of PA6 and its PA6-GF composites were compared both in their unconditioned and hydrothermally aged states in a wet abrasive wear mode at 60 °C. Conditioned samples were kept in demineralized water at different temperatures (25 (RT), 40, and 60 °C) to reach saturation. The tensile mechanical properties of both unconditioned and conditioned specimens of PA6 and its PA6-GF composites were also investigated. Aging in water, in the presence of oxygen, may degrade polyamides due to hydrolysis and oxidation [22]. FTIR was performed to recognize the polymer degradation.

Table 1

The data from fitting curves for water absorption graphs in Fig. 2.

Materials	Temperature (°C)	Fitting equation	Correlation coefficient (R^2)	Diffusion coefficient (D , mm^2/s)
PA6	25 (RT)	$y = 0.1685x$	0.9992	1.782×10^{-6}
PA6-15GF	25 (RT)	$y = 0.1178x$	0.9969	1.544×10^{-6}
PA6-25GF	25 (RT)	$y = 0.0988x$	0.998	1.459×10^{-6}
PA6	40	$y = 0.2898x$	0.9967	6.16×10^{-6}
PA6-15GF	40	$y = 0.2014x$	0.9966	4.97×10^{-6}
PA6-25GF	40	$y = 0.1719x$	0.997	4.87×10^{-6}
PA6	60	$y = 0.5331x$	0.9976	21×10^{-6}
PA6-15GF	60	$y = 0.3697x$	0.9976	17×10^{-6}
PA6-25GF	60	$y = 0.3215x$	0.9977	17×10^{-6}

2. Materials and methods

2.1. Materials

The PA6 (Duralon N6) was used in this study as the base polymer and 13 μm glass fibers were added to make PA6-GF composite materials. The glass fibers have siloxane coating. The PA6 composites contained 15, and 25 wt.% glass fibers.

2.2. Sample preparation

Specimens in a cylindrical shape with a diameter of 10 mm were made using injection molding for wear testing. Also, the ASTM D638 Type V dumbbell-shaped tensile specimens and rectangular sample for dynamic mechanical tests were injection molded using a Babyplast 6/10P with two separate molds. All materials were vacuum dried for 4 h at 80 °C prior to molding. The temperatures of the three heating zones were as follows: plastification: 235 °C, injection chamber: 235 °C, and nozzle: 240 °C. The molds were heated and maintained at 80 °C. The pressure was 30 bar in injection and 40 bar in packing sections on all materials. The parts were allowed to cool down for 20 s before ejection.

2.3. Characterization

The X-ray diffraction (XRD) analysis was carried out using Bruker D8 Advance XRD with a Cu K α radiation source (40 kV, 40 mA) and a measurement at 0.02° step.

Dynamic mechanical analysis (DMA) of rectangular specimens with

the dimensions of 10 × 3 × 50 mm³ was performed on a hybrid rheometer (Discovery HR-2, TA Instrument) using a tree point bending clamp with a temperature sweep from −50 to 150 °C at the heating rate of 5 °C min^{−1}, and the frequency of 1 Hz. The conditioned samples were kept in demineralized water at 60 °C to reach saturation. The water on the sample surface was wiped off before doing the test.

Differential scanning calorimetry (DSC) was used to study the effects of hydrothermal aging on the crystallinity of samples. TA Instruments Discovery DSC 2500 was applied to run the test on 5 mg of each sample. Samples were heated from −40 °C to 240 °C, and heating run was considered to calculate the crystallinity ratio (X_c) using the following equation [23]:

$$X_c = \frac{\Delta H_f}{\Delta H_f^0 \times (1 - W_f)} \quad (1)$$

where ΔH_f is the enthalpy of fusion of the sample, ΔH_f^0 the melting enthalpy for a 100 % crystalline material, taken equal to 240 J/g [24] and W_f the fiber weight fraction.

Friction and wear tests were performed on a pin-on-disk machine (RTEC MFT2000) in which the stationary polymer pin was in contact with Struers SiC paper #320 (grit size: 68 μm) attached to the top of a rotating disk. The diameter of the rotating disk was 50 mm. The pins were abraded against a sandpaper for 5 min to get the pin surface flat and ensure good contact between the pin and sandpaper on the disk. Dry sliding at ambient conditions with a normal load of 15 N and a sliding speed of 250 rpm was performed on a 28 mm diameter wear track. The corresponding wear distance was 220 m. The mass of the pin and coefficient of friction (COF) were measured after 10 min. These parameters were also applied to measure the wear and friction of conditioned samples kept at different temperatures (RT, 40, and 60 °C) once they achieved saturation at demineralized water. The tests were carried out in a container including demineralized water at 60 °C. This condition was selected as 60 °C represents the upper use temperature of PA6 and therefore represents a value of practical significance. The worn surface of pins was studied using SEM Hitachi 3030T.

The tensile testing was performed following the ASTM D638 procedure using an Instron 5584 universal testing machine fitted with a 5 kN load cell, 1 kN pneumatic grips, and an advanced video extensometer

Table 2 T_g of PA6 and its PA6-GF composites.

Materials	T_g (°C)
PA6_Unconditioned	61.9 ± 2.3
PA6-15GF_Unconditioned	57.8 ± 1.5
PA6-25GF_Unconditioned	56.5 ± 0.7
PA6_Conditioned	−11.9 ± 3.5
PA6-15GF_Conditioned	−15.3 ± 0.5
PA6-25GF_Conditioned	−18 ± 0.5

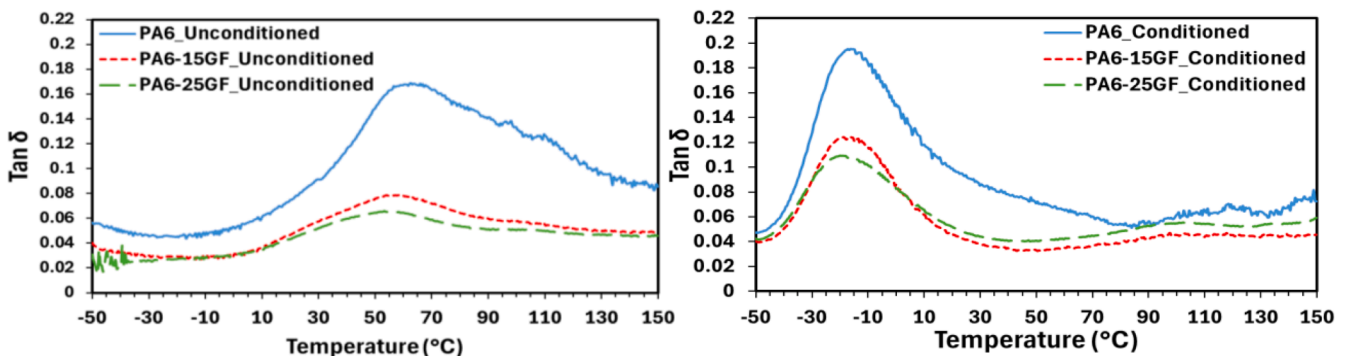


Fig. 3. Tan δ of PA6 and its PA6-GF composites a) without conditioning and b) conditioned at 60 °C.

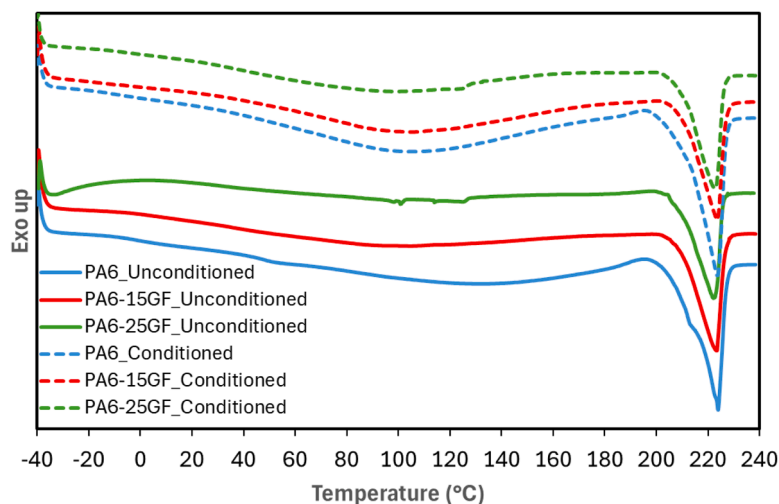


Fig. 4. Heating scans of unconditioned and conditioned (in water at 60 °C) PA6 and PA6-GF composites.

Table 3

Crystallinity ratio of PA6 and its PA6-GF composites.

Materials	X_c (%)
PA6_Unconditioned	27.27 ± 0.18
PA6-15GF_Unconditioned	23.9 ± 0.77
PA6-25GF_Unconditioned	26.62 ± 2.15
PA6_Conditioned	33.05 ± 0.94
PA6-15GF_Conditioned	25.38 ± 0.24
PA6-25GF_Conditioned	29.11 ± 0.37

for strain measurements. The distance between the grips was 25.4 mm, with a testing speed of 10 mm/min. Tests were repeated three times, and data were indicated as mean $\pm$ standard deviation (SD). A one-way ANOVA was used for statistical analysis and differences were significant when $p < 0.05$.

The infrared spectra of unconditioned and conditioned PA6 and PA6-GF were obtained using a Fourier transform infrared (FTIR) spectroscopy (Thermo Scientific, Nicolet iS50)- ATR, with a scanning rate of 32 min^{-1} and the resolution of 2 cm^{-1} in the range of 400–4000 cm^{-1} to determine the functional groups.

2.4. Water testing

Prior to conditioning, samples were dried in the vacuum oven at 80 °C for 4 h. Then samples were immersed in demineralized water to reach saturation at RT, 40, and 60 °C. The mass changes of samples caused by hydrothermal swelling were measured by an analytical balance with an accuracy of 0.1 mg at different time intervals.

3. Results and discussion

3.1. XRD

XRD is used to determine the crystalline structure of materials. As shown in Fig. 1, for neat PA6, the α (monoclinic) crystalline form can be identified by peaks occurring at approximately $2\theta = 20.5^\circ$ and 23° originated from the distance between hydrogen-bonded chains and the distance between hydrogen-bonded sheets, respectively. The peak at approximately $2\theta = 21.3^\circ$ corresponds to the γ (pseudo-hexagonal) crystalline form of PA6 [25–27]. The presence of 15 and 25 wt.% GF in PA6 matrix did not cause obvious changes in XRD diffraction of composites. Literature showed a broad hump ranging from $2\theta = 20^\circ$ to 35° in XRD graphs due to the structural disorder in the glasses [28,29].

3.2. Water absorption behavior

In Fig. 2, the water absorption curves of PA6 and its PA6-GF composites are reported in terms of mass change (%) versus aging time ($\text{h}^{1/2}$) when immersed in demineralized water at the three conditioning temperatures (RT, 40 °C, and 60 °C). The mass changes of the samples can be calculated by the mass difference of specimens before and after water uptake, according to the following equation:

$$W_w(\%) = \frac{W_t - W_d}{W_d} \times 100 \% \quad (2)$$

where W_w , W_t and W_d denote the mass changes, the mass at t time, and the dry mass of samples, respectively. For all three conditions, the same trend of water uptake can be seen in neat PA6 and its composites where water uptake gradually increased and then reached a plateau (equilibrium condition/ saturation moisture content). This absorption behavior is typical of the Fickian absorption mechanism [30–33]. The conditioning time to approach saturation depends on the aging temperature. PA6 reached saturation after 204, 67, and 17 days at RT, 40 °C and 60 °C, respectively. The rate of water uptake speeded up with the higher temperature [34]. At RT, the saturation moisture content of PA6, PA6-15GF, and PA6-25GF are 9.38 %, 7 %, and 6.04 %, respectively. The equilibrium saturation percentage however, declined, for example, at 40 °C, it is 8.65 %, 6.67 %, and 5.75 % for PA6, PA6-15GF, and PA6-25GF, respectively, which are in agreement with literature data [16].

The water absorption following Fickian diffusion can be characterized by a constant diffusion coefficient (D) and an equilibrium saturation mass (W_∞). In the initial process of water absorption, before approaching equilibrium, the graphs in Fig. 2 show a linear relationship as per the following equation [35,36]:

$$\frac{W_w}{W_\infty} = \frac{4}{\sqrt{\pi}} \sqrt{\frac{Dt}{L^2}} \quad (3)$$

where L is the sample thickness. The slope of the linear part of the water absorption curve (K) can be calculated through the fitting curve. Then, the D was determined through the following equation:

$$D = \frac{\pi K^2 L^2}{16 W_\infty^2} \quad (4)$$

Fitting straight-line equations for the diffusion of water of PA6 and PA6-GF composites in Fig. 2 as a function of hydrothermal aging time at different temperatures are shown in Table 1. These values agree well with literature values [16]. Consistent with expectations [37], the diffusion coefficient decreased with increasing fiber loading and the

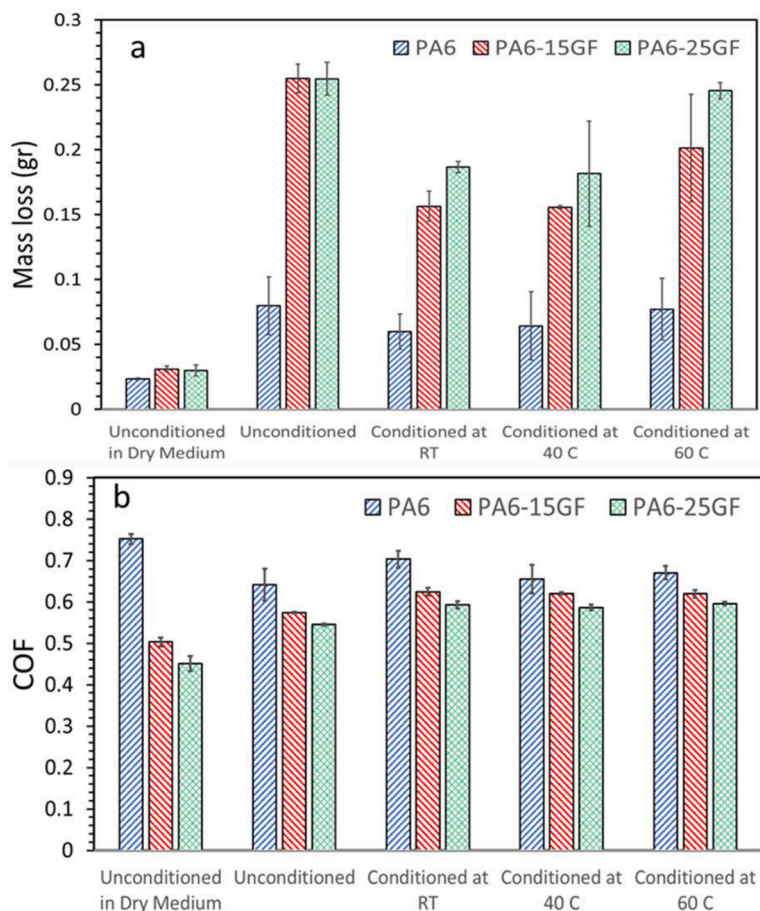


Fig. 5. Wear behavior, a) Mass loss and b) COF of unconditioned and conditioned PA6 and its PA6-GF composites exposed to water at different temperatures (RT, 40, and 60 °C) in abrasion pin-on-disk tests in water medium at 60 °C, unconditioned samples was also tested in a dry medium as a reference.

diffusion coefficient increased with increasing the water temperature.

3.3. Effect of hydrothermal aging on T_g and crystallinity

Dynamic mechanical analysis was performed within a temperature sweep from -50 to 150 °C at the frequency of 1 Hz to determine the $\tan \delta$ of PA6 and PA6-GF composite specimens. The conditioned samples were kept at 60 °C and tested at the saturation state. $\tan \delta$ or damping factor, is the ratio of loss modulus to storage modulus representing the degree of molecular mobility in the polymer matrices. The glass transition temperature (T_g) is estimated at the temperature where $\tan \delta$ peaks. The variation of $\tan \delta$ as a function of temperature is shown in Fig. 3a and b and the corresponding T_g is shown in Table 2. $\tan \delta$ of unconditioned and conditioned samples decreased with the addition of glass fiber indicating the fibers limited the mobility of the polymer chains. In both unconditioned and conditioned specimens, the presence of glass fiber marginally decreased the T_g [38,39]. In conditioned samples, T_g was depressed because of an increase in the mobility of PA6 chains (plasticization), and swelling phenomena by the absorbed water. Consistent with the literature, hydrothermal aging results in a decrease in the T_g value of polyamide composites [16,36]. During the aging of polyamides, water is mainly absorbed in the amorphous regions and T_g reduction occurs through the mechanism of the replacement of intercatenary hydrogen bonds in the accessible amide groups by interaction with water [40]. The negative T_g of conditioned samples shows that PA6 and PA6-GF composites turned into a rubbery state due to hydrothermal conditioning in this work [23].

The heating graphs of DSC are shown in Fig. 4. Conditioned specimens were kept at 60 °C and tested once reached saturation state. The

mass of residual water was considered in calculating the crystallinity ratio of conditioned samples. Results show that hydrothermal aging led to a slight increase in the enthalpy of the melting peak of PA6. Table 3 shows the crystallinity ratio of PA6 increased by around 6 % after hydrothermal aging at 60 °C and 2 months, which can be associated with the chemi-crystallization phenomenon. Hydrothermal aging can cause molecular chain scission and molecule segments in the amorphous region can form new crystalline structures if they have enough mobility [23,41]. Upon hydrothermal aging, PA6-GF composite samples did not show significant changes in the crystallinity ratio.

3.4. Pin-on-disk test

In this study, the unconditioned and conditioned PA6 and PA6-GF composite samples were tribo-tested in an abrasive mode in demineralized water at 60 °C. As explained in Section 2, the samples were conditioned at three temperature levels (RT, 40, and 60 °C) and tested once they reached saturation. The pin-on-disk tests were also carried out on as-received pins in a dry medium as a reference. A comparison between the abrasion mass loss and COF of unconditioned and conditioned PA6 and its PA6-GF composites samples is shown in Fig. 5. As can be seen in Fig. 5, a surprising finding is that there is no statistically significant difference in mass loss between the unconditioned and conditioned neat PA6 samples when tested in water medium. These results highlight that conditioning PA6 samples at different temperatures up to saturation state; does not affect the wear resistance of the polymer. It may also come as a surprise that conditioned PA6-GF composites showed less mass loss compared to unconditioned composites. Similar findings were found by Guerrazi et al. [4] when they compared the

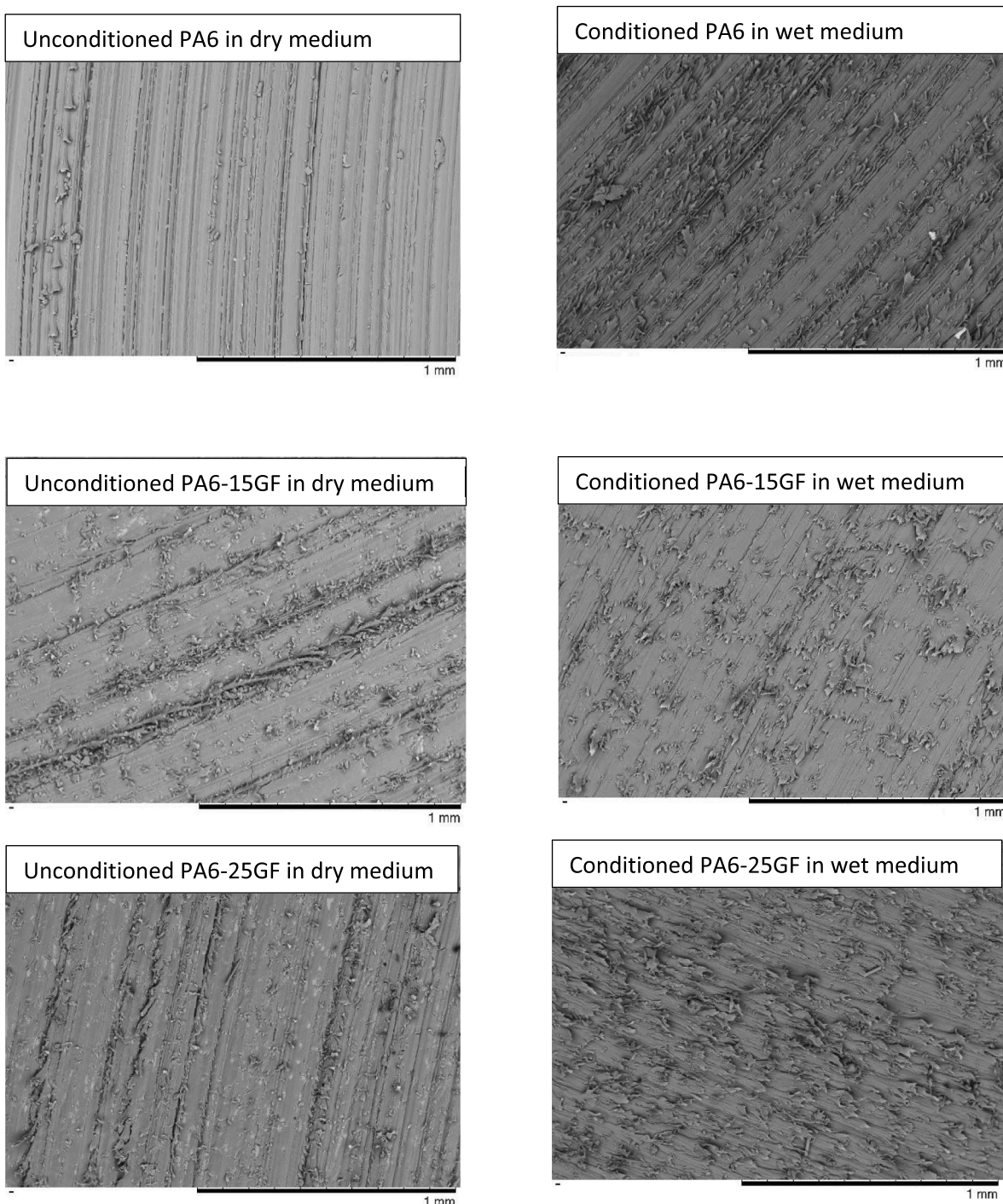


Fig. 6. SEM morphologies of worn surfaces of unconditioned samples tested in dry medium (left column) and conditioned samples at 60 °C tested in wet medium (right column).

abrasion wear tests between unconditioned and conditioned glass fiber reinforced epoxy at saturation state. There was noticeably less wear (mass loss) where the abrasion tests were carried out in a dry medium. It is assumed that for samples conducted in water, the transfer film of worn materials deposited on the sandpaper was washed away, and therefore more aggressive wear conditions were created. In all tests, the existence of glass fibers led to reducing the wear resistance. This was more pronounced for tests performed in a water medium compared to those performed dry. The SEM images in Fig. 6 show the differences of worn surfaces in dry and wet media. It can be seen that the worn surfaces of conditioned samples display more ploughed marks and exfoliating

regions indicating more wear mass loss. It may be attributed to the plasticization and softening which occur through water diffusion into the polymer structure [42].

In all tests, the existence of glass fibers caused a reduction in the COF compared to the neat PA6. As shown in Fig. 5b, the average COF of neat PA6 was higher in the dry medium than in the wet tests. This suggests that water acts as a lubricant in this test configuration [43]. As explained above, the water washed away residues that formed the transfer layer. However, this trend changed for PA6-GF composites. The COF of conditioned PA6-GF composites specimens were marginally higher than the unconditioned samples, which can be attributed to the diffusion of

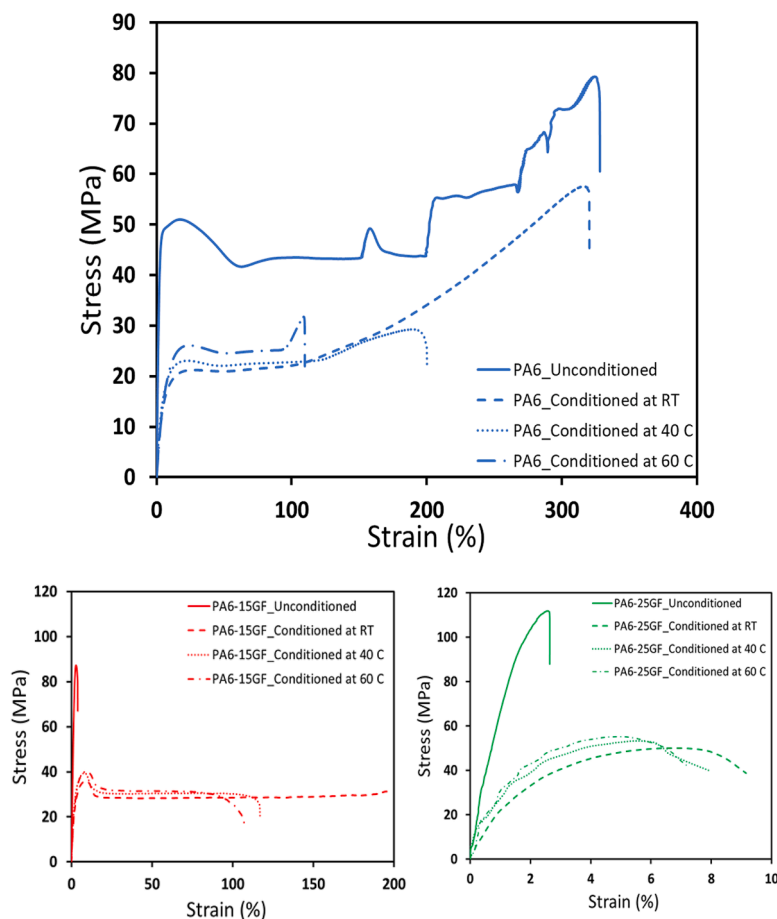


Fig. 7. Stress-strain graphs of unconditioned and conditioned specimens made of PA6 and PA6-GF composites at RT, 40 °C and 60 °C.

water molecules into the polymer matrix which leads to plasticization phenomena of polymer matrix [44]. Aging in water caused conditioned samples to turn into a rubbery state and remain in the same state at 60 °C (abrasion wear test temperature). Subsequently, the degradation in mechanical properties may happen upon hydrothermal aging, for example, reductions in the elastic modulus of polymer structure which results in an increase in the contact area and higher COF [42,43]. It is not easy to predict the wear rate of polymers because of variations in wear mechanisms depending on operation conditions and material properties. A wear model was suggested by Ratner and Lancaster [45] representing a relation between the wear volume of polymers and their mechanical properties where the wear volume is inversely proportional to hardness, tensile strength, and strain at break [46]. Thus, the impact of hydrothermal aging on the mechanical behavior of PA6 and its composites will be assessed in the next section.

3.5. Tensile testing

Fig. 7 presents the stress-strain graphs of unconditioned and conditioned specimens at different temperatures. The corresponding data of tensile testing are presented in Fig. 8, which are consistent with the previous studies [16]. In both unconditioned and conditioned specimens, the presence of glass fiber generally can lead to a larger increase in the tensile modulus of PA6 matrix compared to the maximum tensile strength. For example, in unconditioned samples, the tensile modulus of PA6 rose by 108 % and 176 % with the addition of 15 % GF and 25 % GF to the polymer matrix, respectively. However, the maximum tensile strength of PA6 matrix increased by only 14 % and 40 % with the addition of 15 % GF and 25 % GF, respectively.

On the other hand, conditioning caused a drop in both the tensile modulus and maximum tensile strength of all samples. The reduction in

tensile modulus of PA6 specimens was higher (~83 %) compared to PA6-15GF composites (~61 %) and PA6-25GF composites (~52 %), showing that the bonding between glass fibers and polymer chains could boost the stiffness of matrix, even though the polymer matrix softening happened. Reduction in modulus of samples under hydrothermal aging can be attributed to the reduction in T_g [36]. In this work, T_g of specimens was dropped by 70 °C after aging in water at 60 °C. During aging, polyamides absorb water through the replacement of intercatenary hydrogen bonds in accessible amide groups which causes plasticization, leading to an increase in the bond length and a decrease in the Van-derwaals force [37,40]. The maximum tensile strength of PA6 and PA6-GF specimens under conditioning were decreased by ~50–60 % compared to unconditioned samples.

Fig. 7 displays the presence of glass fibers led to less strain at break. The same behavior was found for glass fiber-polybutylene terephthalate (PBT) composite [47]. However, changes in the strain at break according to water uptake were different for PA6 and PA6-GF specimens. Conditioned samples of PA6 demonstrated less strain at break compared to unconditioned samples, while water absorption in PA6-GF composites caused higher strain at break compared to unconditioned samples. It can be attributed to different patterns of water diffusion into neat PA6 and PA6-GF composites [4]. Further, it should be noted that conditioning increased the crystallinity ratio of neat PA6 by around 6 % which may be due to chemi-crystallization phenomena, representing molecular chain scission. Strain at break is the most relevant mechanical characterization parameter to recognize embrittlement induced by chemical degradation during hydrothermal aging. Strain at break significantly decreased once PA6 was exposed to water at elevated temperature (60 °C), from 391 % to 95 %, which can be associated with embrittlement mechanism and hydrolysis. On the other hand, strain at break for PA6-GF composites

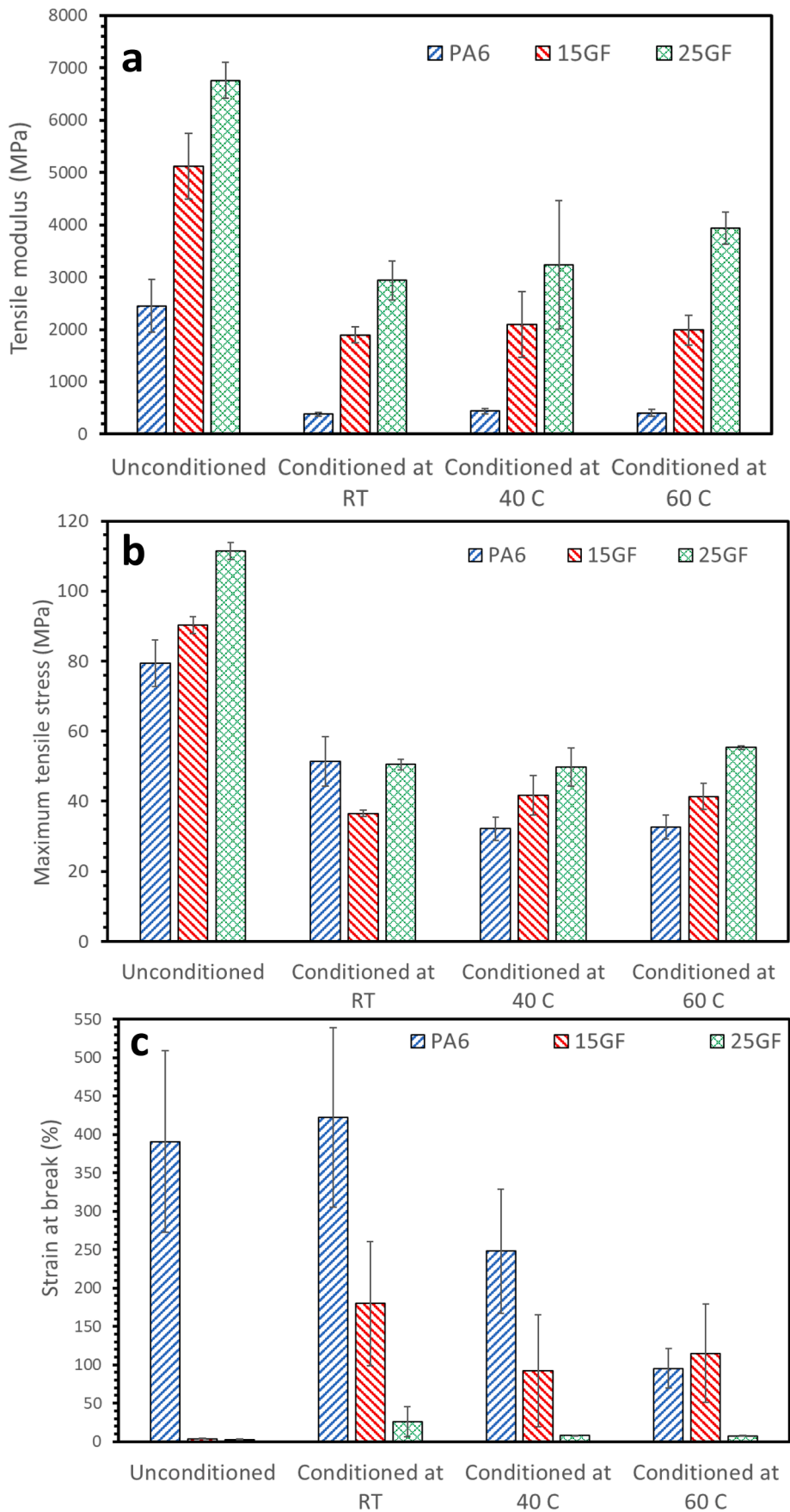


Fig. 8. a) Tensile modulus, b) maximum tensile stress, and c) strain at break of unconditioned and conditioned specimens at RT, 40 °C, and 60 °C made of PA6 and PA6-GF composites.

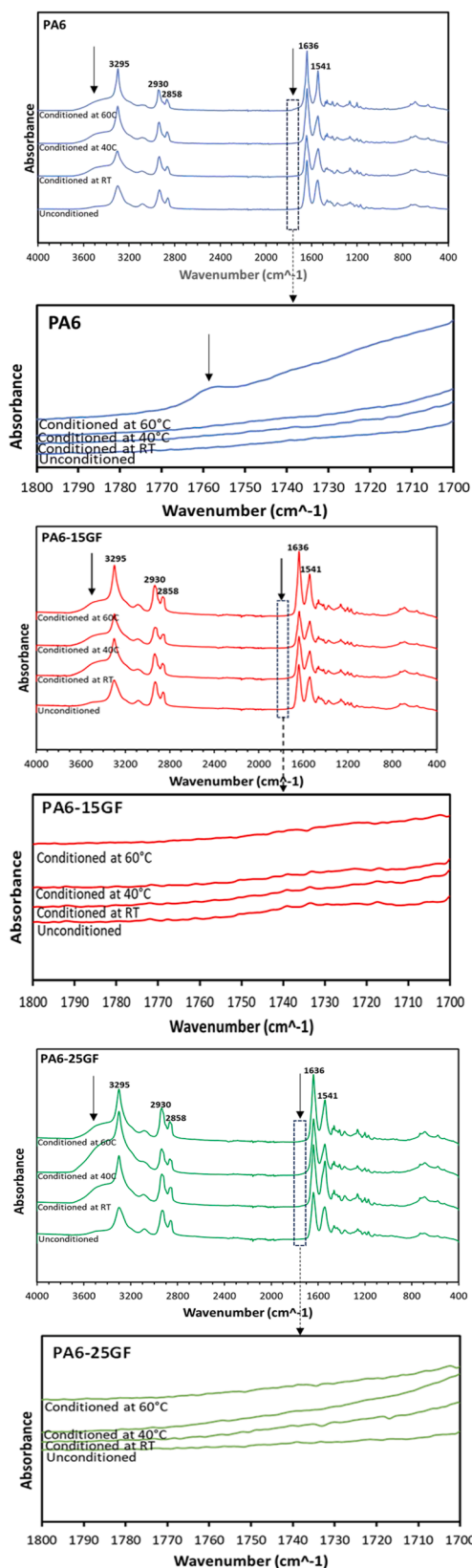


Fig. 9. FTIR spectra of unconditioned and conditioned PA6, PA6-15GF, and PA6-25GF.

increased when they were under hydrothermal aging. In this work, all samples were tested at saturation state, however, literature has shown that longer immersion time can cause a significant reduction in the strain at break of unfilled polyamide. They reported a sudden drop in strain at break once samples were exposed to water at 80 °C between 3 and 6 months while the glass fiber polyamide composites showed a more gradual decrease [48].

As reported wear rate of the polymer is inversely proportional to the combination of strength and strain at break [46]. Accordingly, the abrasive wear behavior of PA6-GF composites in this study may be explained as contributing to a combination of mechanical variables. The mass loss of conditioned PA6-GF specimens was less than that of the unconditioned ones, possibly due to, a reduction in tensile strength and an increase in strain at break.

3.6. FTIR tests

FTIR analysis was performed to determine the differences in chemical compositions of unconditioned and conditioned samples. Fig. 9 shows the FTIR spectra of conditioned and unconditioned PA6 and PA6-GF composites' materials. The major peaks at all samples were observed at 3295 cm^{-1} which is assigned to N—H stretching bands. The peaks at 2930 and 2858 cm^{-1} are linked to the stretching C—H. The vibrational band observed at 1636 cm^{-1} refers to Amide I, and the peaks at 1541 cm^{-1} are due to Amide II [37,49,50]. For all conditioned samples, there was a rise in graphs at around 3500 cm^{-1} , representing hydrogen bonded symmetric O—H, which is likely due to water molecules [51]. The absorption peak intensities in the region of 1650–1785 cm^{-1} can be assigned to an increased abundance of carbonyl (C=O), carboxyl (COOH), and ester (COOR) groups [49]. Unfilled PA6 aged at 60 °C shows an increasing peak intensity at 1757 cm^{-1} which can represent polymer oxidation. Deshoules et al. [24] reported an oxidative degradation of PA6 aged in oxygenated water at 80 °C. Signals observed at 1756 and 1734 cm^{-1} were assigned to C=O of isolated carboxylic acids and C=O of imides, respectively. As the detection of hydrolysis products is not easy using FTIR, more future quantitative characterization will need to be performed to measure the molar mass changes.

4. Conclusions

Although mechanisms and consequences of water absorption have been studied for certain PAs, there are still several open questions regarding the relations between the parameters of hydrothermal aging of PAs and their abrasion behavior. The abrasion wear resistance and tensile mechanical properties of unconditioned and conditioned PA6 and PA6-GF specimens were assessed in this paper. The durability of these composites was analysed for polymers bearing used in an aggressive environment (slurry up to 60 °C). Samples conditioning was performed in demineralized water at RT, 40, and 60 °C. Our data indicated that temperature could accelerate water absorption in all samples, and glass fibers in polymer matrix reduced the water uptake. The presence of up to 25 wt.% glass fiber in PA6-based composites exhibits an increasing trend in abrasion wear mass loss and a reduction in COF. Compared to unconditioned specimens, conditioning samples up to 60 °C and allowing them to reach saturation states did not reduce the abrasion wear resistance where tests were performed in a wet medium. Hydrothermal aging of specimens reduced the tensile modulus and strength of unfilled PA6 and PA6-GF composites. However, increasing the temperature up to 60 °C reduced the strain at the break of unfilled PA6, and increased elongation of PA6-GF composites. The reduction in strain at the break of unfilled PA6 at 60 °C can be attributed to chemical degradation which is also shown in the FTIR graph. This study showed that the abrasive wear behavior of PA6-GF composites may be correlated with their mechanical properties during hydrothermal aging.

CRediT authorship contribution statement

Hadis Khakbaz: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Asanka Pradeep Basnayake:** Writing – review & editing, Methodology, Conceptualization. **Akshay Krishna Ambika Harikumar:** Methodology. **Mahshid Firouzi:** Writing – review & editing, Conceptualization. **Darren Martin:** Writing – review & editing, Conceptualization. **Michael Heitzmann:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Michael Heitzmann reports financial support was provided by Australian Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] K. Friedrich, Z. Zhang, A.K. Schlarb, Effects of various fillers on the sliding wear of polymer composites, *Compos. Sci. Technol.* 65 (2005) 2329–2343.
- [2] K. Friedrich, Polymer composites for tribological applications, *Adv. Ind. Eng. Polym. Res.* 1 (2018) 3–39.
- [3] A. Bahrololoumi, V. Morovati, M. Shaafaei, R. Dargazany, A multi-physics approach on modeling of hygrothermal aging and its effects on constitutive behavior of cross-linked polymers, *J. Mech. Phys. Solids* 156 (2021) 104614.
- [4] N. Guermazi, A. Ben Tarjem, I. Ksouri, H.F. Ayedi, On the durability of FRP composites for aircraft structures in hygrothermal conditioning, *Compos. Part B* 85 (2016) 294–304.
- [5] S. Alessi, G. Pitarresi, G. Spadaro, Effect of hydrothermal ageing on the thermal and delamination fracture behaviour of CFRP composites, *Compos. Part B* 67 (2014) 145–153.
- [6] L. Mansouri, A. Djebbar, S. Khatir, M. Abdel Wahab, Effect of hygrothermal aging in distilled and saline water on the mechanical behaviour of mixed short fibre/woven composites, *Compos. Struct.* 207 (2019) 816–825.
- [7] I. Carrascal, J. Casado, J. Polanco, F. Gutiérrez-Solana, Absorption and diffusion of humidity in fiberglass-reinforced polyamide, *Polym. Compos.* 26 (2005) 580–586.
- [8] C.J. Tsenoglou, S. Pavlidou, C.D. Papaspyrides, Evaluation of interfacial relaxation due to water absorption in fiber–polymer composites, *Compos. Sci. Technol.* 66 (2006) 2855–2864.
- [9] A. Harsha, U. Tewari, The effect of fibre reinforcement and solid lubricants on abrasive wear behavior of polyetheretherketone composites, *J. Reinf. Plast. Compos.* 22 (2003) 751–767.
- [10] G. Zhao, I. Hussainova, M. Antonov, Q. Wang, T. Wang, Friction and wear of fiber reinforced polyimide composites, *Wear* 301 (2013) 122–129.
- [11] H.H. Parikh, P.P. Gohil, Tribology of fiber reinforced polymer matrix composites—A review, *J. Reinf. Plast. Compos.* 34 (2015) 1340–1346.
- [12] J. Bijwe, J. Indumathi, J.J. Rajesh, M. Fahim, Friction and wear behavior of polyetherimide composites in various wear modes, *Wear* 249 (2001) 715–726.
- [13] Y. Tokiwa, B.P. Calabria, C.U. Ugwu, S. Aiba, Biodegradability of plastics, *Int. J. Mol. Sci.* 10 (2009) 3722–3742.
- [14] S. Kumar, K. Panneerselvam, Two-body abrasive wear behavior of nylon 6 and glass fiber reinforced (GFR) nylon 6 composite, *Proc. Technol.* 25 (2016) 1129–1136.
- [15] A.K. Ambika Harikumar, M.T. Heitzmann, A. Basnayake, H. Khakbaz, D. Martin, Tribological performance of nylon-6/oyster shell composites, in: *ASME 2023 42nd International Conference on Ocean, Offshore and Arctic Engineering*, 2023.
- [16] I. Ksouri, N. Haddar, Long term ageing of polyamide 6 and polyamide 6 reinforced with 30 % of glass fibers: temperature effect, *J. Polym. Res.* 25 (2018) 1–12.
- [17] T. Sathishkumar, S. Satheshkumar, J. Naveen, Glass fiber-reinforced polymer composites—A review, *J. Reinf. Plast. Compos.* 33 (2014) 1258–1275.
- [18] C.J. Hooke, S.N. Kukureka, P. Liao, M. Rao, Y.K. Chen, Wear and friction of nylon-glass fibre composites in non-conformal contact under combined rolling and sliding, *Wear* 197 (1996) 115–122.
- [19] H.C.Y. Cartledge, C. Baillie, Y.-W. Mai, Friction and wear mechanisms of a thermoplastic composite GF/PA6 subjected to different thermal histories, *Wear* 194 (1996) 178–184.
- [20] S.S. Kim, M.W. Shin, H. Jang, Tribological properties of short glass fiber reinforced polyamide 12 sliding on medium carbon steel, *Wear* 274 (2012) 34–42.
- [21] D. Ferreño, I. Carrascal, E. Ruiz, J.A. Casado, Characterisation by means of a finite element model of the influence of moisture content on the mechanical and fracture properties of the polyamide 6 reinforced with short glass fibre, *Polym. Test.* 30 (2011) 420–428.
- [22] Q. Deshouilles, M. Le Gall, C. Dreanno, M. Arhant, D. Priour, P.Y. Le Gac, Chemical coupling between oxidation and hydrolysis in polyamide 6—A key aspect in the understanding of microplastic formation, *Polym. Degrad. Stab.* 197 (2022) 109851.
- [23] M. Arhant, E. Lolive, T. Bonnemains, P. Davies, A study of pure hydrolysis of carbon fibre reinforced polyamide 6 composites tested under mode I loading, *Compos. Part A* 152 (2022) 106719.
- [24] Q. Deshouilles, M. Le Gall, C. Dreanno, M. Arhant, D. Priour, P.Y. Le Gac, Modelling pure polyamide 6 hydrolysis: influence of water content in the amorphous phase, *Polym. Degrad. Stab.* 183 (2021) 109435.
- [25] T. Fornes, D.R. Paul, Crystallization behavior of nylon 6 nanocomposites, *Polymer* 44 (2003) 3945–3961.
- [26] X. Hu, X. Zhao, Effects of annealing (solid and melt) on the time evolution of polymorphic structure of PA6/silicate nanocomposites, *Polymer* 45 (2004) 3819–3825.
- [27] J.C. Farias-Aguilar, M.J. Ramírez-Moreno, L. Téllez-Jurado, H. Balmori-Ramírez, Low pressure and low temperature synthesis of polyamide-6 (PA6) using NaO as catalyst, *Mater. Lett.* 136 (2014) 388–392.
- [28] G. Lakshminarayana, K.M. Kaky, S. Baki, A. Lira, P. Nayar, I. Kityk, M. Mahdi, Physical, structural, thermal, and optical spectroscopy studies of $\text{TeO}_2\text{--B}_2\text{O}_3\text{--MoO}_3\text{--ZnO--R}_2\text{O}$ ($\text{R} = \text{Li, Na, and K}$)/ MO ($\text{M} = \text{Mg, Ca, and Pb}$) glasses, *J. Alloys Compd.* 690 (2017) 799–816.
- [29] J. Liu, M. Chen, J. Yang, Z. Wu, Study on mechanical properties of basalt fibers superior to E-glass fibers, *J. Nat. Fibers* 19 (2022) 882–894.
- [30] H. Ardebili, J. Zhang, M.G. Pecht, 6—Characterization of encapsulant properties, in: H. Ardebili, J. Zhang, M.G. Pecht (Eds.), *Encapsulation Technologies for Electronic Applications*, second ed., William Andrew Publishing, 2019, pp. 221–258.
- [31] V. Prasad, M.A. Joseph, K. Sekar, Investigation of mechanical, thermal and water absorption properties of flax fibre reinforced epoxy composite with nano TiO_2 addition, *Compos. Part A* 115 (2018) 360–370.
- [32] A. Ladhari, H. Ben Daly, H. Belhadjsalah, K.C. Cole, J. Denault, Investigation of water absorption in clay-reinforced polypropylene nanocomposites, *Polym. Degrad. Stab.* 95 (2010) 429–439.
- [33] R.K. Nayak, B.C. Ray, Water absorption, residual mechanical and thermal properties of hydrothermally conditioned nano- Al_2O_3 enhanced glass fiber reinforced polymer composites, *Poly. Bull.* 74 (2017) 4175–4194.
- [34] H. Wang, G. Xian, H. Li, L. Sui, Durability study of a ramie-fiber reinforced phenolic composite subjected to water immersion, *Fibers Polym* 15 (2014) 1029–1034.
- [35] R. Li, L. Ye, G. Li, Long-term hydrothermal aging behavior and aging mechanism of glass fibre reinforced polyamide 6 composites, *J. Macromol. Sci., Part B* 57 (2018) 67–82.
- [36] V. Venoor, J.H. Park, D.O. Kazmer, M.J. Sobkowicz, Understanding the effect of water in polyamides: a review, *Polym. Rev.* 61 (2021) 598–645.
- [37] Y. Lei, T. Zhang, J. Zhang, B. Zhang, Dimensional stability and mechanical performance evolution of continuous carbon fiber reinforced polyamide 6 composites under hygrothermal environment, *J. Mater. Res. Technol.* 13 (2021) 2126–2137.
- [38] M.H. Liu, R. Li, G. Wang, Z.Y. Hou, B. Huang, Morphology and dynamic mechanical properties of long glass fiber-reinforced polyamide 6 composites, *J. Therm Anal Calorim* 126 (2016) 1281–1288.
- [39] W. Feng, A. Ait-Kadi, J. Brisson, B. Riedl, Polymerization compounding composites of nylon-6, 6/short glass fiber, *Polym. Compos.* 24 (2003) 512–524.
- [40] H. Reimschuessel, Relationships on the effect of water on glass transition temperature and young's modulus of nylon 6, *J. Polym. Sci.* 16 (1978) 1229–1236.
- [41] M.S. Rabello, J.R. White, Crystallization and melting behaviour of photodegraded polypropylene—I. Chemo-crystallization, *Polymer* 38 (1997) 6379–6387.
- [42] N. Guermazi, K. Elleuch, H.F. Ayedi, P. Kapsa, Aging effect on thermal, mechanical and tribological behaviour of polymeric coatings used for pipeline application, *J. Mater. Process. Technol.* 203 (2008) 404–410.
- [43] H. Meng, G.X. Sui, G.Y. Xie, R. Yang, Friction and wear behavior of carbon nanotubes reinforced polyamide 6 composites under dry sliding and water lubricated condition, *Compos. Sci. Technol.* 69 (2009) 606–611.
- [44] A.K. Bandaru, P.M. Weaver, R.M. O'Higgins, Abrasive wear performance of hygrothermally aged glass/PTFE composites, *Polym. Test.* 103 (2021) 107369.
- [45] J. Lancaster, Relationships between the wear of polymers and their mechanical properties, in: *Proceedings of the Institution of Mechanical Engineers, Conference Proceedings*, SAGE Publications Sage UK, London, England, 1968.

- [46] F. Hakami, A. Pramanik, A.K. Basak, N. Ridgway, Elastomers' wear: comparison of theory with experiment, *Tribol. Int.* 135 (2019) 46–54.
- [47] O. Hamlaoui, O. Klinkova, R. Elleuch, I. Tawfiq, Effect of the glass fiber content of a polybutylene terephthalate reinforced composite structure on physical and mechanical characteristics, *Polymers* 14 (2022) 17.
- [48] P.-Y. Le Gac, B. Fayolle, Impact of fillers (short glass fibers and rubber) on the hydrolysis-induced embrittlement of polyamide 6.6, *Compos. Part B* 153 (2018) 256–263.
- [49] D. Pavliňák, J. Hnilica, A. Quade, J. Schäfer, M. Alberti, V. Kudrle, Functionalisation and pore size control of electrospun PA6 nanofibres using a microwave jet plasma, *Polym. Degrad. Stab.* 108 (2014) 48–55.
- [50] G. Rotter, H. Ishida, FTIR separation of nylon-6 chain conformations: clarification of the mesomorphous and γ -crystalline phases, *J. Polym. Sci. B* 30 (1992) 489–495.
- [51] M.M. Brette, A.H. Holm, A.D. Drozdov, J. De Claville Christiansen, Pure hydrolysis of polyamides: a comparative study, *Chemistry* 6 (2023) 13–50.